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Third International Conference on Networking and Communications : Submission (2235) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> 22 February 2026 at 23:03
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Hello,

The following submission has been created.

Track Name: ICNWC2026

Paper ID: 2235

Paper Title: Actionable Risk Stratification: A Counterfactual XAI Approach to Lifestyle-Induced NCD Management via Probabilistic Multimodal Fusion

Abstract:
Non-communicable diseases (NCDs) such as diabetes, heart disease, and chronic kidney disease account for a large share of deaths worldwide and continue to put heavy pressure on healthcare systems. Preventive healthcare approaches are
becoming increasingly important, but current diagnostic methods
often fail to combine lifestyle information with clinical data
effectively, and many machine learning models lack transparency
for practical clinical use. In this work, we propose an Integrated
Explainable Artificial Intelligence (XAI) framework to support
early risk assessment of lifestyle-related NCDs. The proposed
system combines clinical indicators such as blood glucose and
blood pressure with lifestyle factors including physical activity
and dietary habits through a multimodal data fusion approach.
To improve prediction performance, an adaptive ensemble model
combining XGBoost, Random Forest, and Logistic Regression is
used, where model contributions are dynamically adjusted based
on performance. To make predictions more useful for practical
decision-making, the framework also includes a counterfactual
explanation module based on SHAP values, which helps generate
actionable lifestyle recommendations. For example, the system
can estimate how changes such as reducing BMI may lower
disease risk. Experiments conducted on a combined dataset show
that the proposed model achieves an accuracy of 96.2% and an
F1-score of 96.1%, outperforming individual baseline models.
Overall, the framework provides accurate predictions while
also offering interpretable insights that can support preventive
healthcare decisions.

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Authors:
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Secondary Subject Areas: Not Entered

Submission Files:
NEW MAJOR FINAL PAPER.pdf (319 Kb, Sun, 22 Feb 2026 16:42:04 GMT)

Submission Questions Response: Not Entered

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